

ASSIGNMENT BRIEF

HTU Course No: 40201450	HTU Course Name: Cloud Computing
BTEC Unit Code:	BTEC UNIT Name:



Student Name/ID Number/Section	
HTU Course Number and Title	40201450 Cloud Computing
BTEC Unit Code and Title	
Academic Year	2025-2026 2
Assignment Author	Yazan Alshannik
Course Tutor	Yazan Alshannik
Assignment Title	Cloud Computing Systems Design and Implementation Project
Assignment Ref No	1
Issue Date	27/04/2026
Formative Assessment dates	From 27/04/2026 to 04/06/2026
Submission Date	13/06/2026
IV Name & Date	Qutaiba Albluwi 26/04/2026
Submission Format	

Submission Instructions

The assignment must be submitted as a soft copy through the university's eLearning platform before the specified deadline using the following link:

<https://elearning.htu.edu.jo>

Your submission must include:

- A Word document report (without borders)
- All required supporting files
- A completed and properly signed Student Declaration Form

Submission Requirements – Part 1

Your submission must include an individual written report that follows the guidelines provided in this document and addresses all sections of the assignment brief.

Report Guidelines

Your written report must meet the following standards:

- The report must be submitted as a Word document
- It must be written in a formal business style
- Font size: 12
- Line spacing: Single
- All answers must clearly indicate the task number and sub-task number
- You must provide a step-by-step explanation for every action performed
- Your report must include properly formatted references using the Harvard Referencing System

- The report must address both scenarios provided in the assignment brief. Completion of only one scenario will be considered incomplete.
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Submission Rules

- All submissions must be uploaded to the eLearning platform before the deadline
 - Email submissions will not be accepted
 - Late submissions will not be accepted under any circumstances
 - Any instance of plagiarism will result in penalties according to HTU policies
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Use of AI Tools

The use of AI tools is allowed in a limited and responsible manner. However, you must:

- Fully understand all generated content
 - Carefully review, edit, and adapt it yourself
 - Clearly mention in your report that AI tools were used
 - Explain how AI tools contributed to your work
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In-Class Examination

An in-class written examination will be conducted prior to the oral examination. This exam will be closed book and notes, and students are expected to provide clear, well-structured, and detailed answers that demonstrate a solid understanding of cloud computing concepts and their practical applications.

The exam may cover topics such as:

- Fundamental cloud computing concepts, including service models and the key benefits of cloud adoption
- The structure of global cloud infrastructure, including regions, availability zones, and edge locations
- The differences between traditional infrastructure, virtualization, and container-based environments, and how each approach impacts performance, scalability, and resource utilization
- The role of cloud architecture in improving system availability, scalability, and fault tolerance
- A comparison between server-based and serverless architectures, including their trade-offs in terms of cost, scalability, and operational complexity
- An overview of cloud database solutions and an introduction to machine learning services available in cloud platforms

Students may also be required to analyze real-world scenarios and justify their technical decisions.

Oral Examination (Viva)

An in-class oral examination will be conducted after the submission deadline.

Students are required to:

- Provide clear, detailed, and technically accurate answers
- Demonstrate full understanding of their submission

The oral exam may include:

- Questions related to course topics and assessment criteria
- Deployment of applications on cloud platforms
- Explanation of cloud services
- Troubleshooting real-world scenarios

Failure to demonstrate understanding during the oral examination may affect the final grade.

Unit Learning Outcomes

LO1 Understand Cloud Concepts and Global Infrastructure

LO2 Manage cloud Identity, Security, and Networking

LO3 Work with Core cloud Compute, Storage, and Database Services

LO4 Integrate and Build Cloud Applications

LO5 Deploy, Optimize, and Operate Cloud Systems

Assignment Brief and Guidance

Scenario 1: HTUNews Cloud Migration

This is Yazan from HTUNews, a startup news company delivering content across the Middle East. The company is currently operating entirely on an on-premise infrastructure. However, with the recent growth in users and traffic, several operational and scalability challenges have started to emerge.

As a result, HTUNews has decided to migrate its infrastructure to Amazon Web Services (AWS). The decision to adopt cloud computing has already been made; therefore, the focus of this engagement is not on explaining its benefits, but rather on designing and implementing appropriate cloud-based solutions.

To ensure a smooth and controlled transition, the migration will be executed in phases, rather than a full immediate shift. This phased approach will allow the gradual migration of system components to the cloud while maintaining system stability and minimizing risk. Accordingly, multiple iterations and continuous improvements are expected throughout the migration process.

Access Management and Security

This is Yazan from HTUNews, a startup news company delivering content across the Middle East. The company is currently operating entirely on an on-premise infrastructure. However, with the recent growth in users and traffic, several operational and scalability challenges have started to emerge.

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The company is structured into several departments, as follows:

Department	Employees
Admins	Yazan, Ahmad
Developers	Lina, Omar, Khaled
Content Editors	Sara, Noor, Huda, Ali

Table 1: Company Departments and Employees

Each department has different responsibilities within the system:

Department	Responsibilities
Admins	Responsible for managing the entire infrastructure, including provisioning servers, managing storage, configuring policies, and monitoring the system.
Developers	Responsible for deploying and updating the application, and working with application-related servers, without requiring full access to all infrastructure resources or sensitive security configurations.
Content Editors	Responsible only for uploading and managing news content, including images and media, without access to server configurations or infrastructure settings.

Table2: Department Roles and Responsibilities

To maintain secure and efficient access control, each user should be granted only the permissions required to perform their assigned responsibilities, without interfering with other areas of the system. Permissions should be organized in a structured and manageable manner through role-based grouping, rather than being assigned directly to individual users. In addition, sensitive accounts, particularly administrative accounts, must be protected with an additional layer of security to reduce the risk of unauthorized access.

Based on the above structure, HTUNews need you to design and explain a proper Identity and Access Management (IAM) model, including how users, groups, and permissions should be organized to ensure secure and controlled access to our cloud resources.

Availability and Scalability

At this stage, the platform handles approximately 10,000 users per hour under normal conditions. However, during breaking news events, traffic may surge significantly, reaching up to 80,000 users within a single hour. Such spikes place substantial pressure on the existing servers and may lead to service degradation or downtime if not managed properly.

To address this, an appropriate server type must be selected based on current demand and anticipated future growth. The selected solution should provide a balance between performance, cost efficiency, and scalability. The application currently runs on a Linux-based environment and is exposed on port 8080. In addition, secure SSH access may be required for administration and maintenance purposes.

High availability is also a critical requirement. The website should remain accessible even if one of the servers fails. For this reason, a health check mechanism is required to continuously monitor application availability and responsiveness. Moreover, if a server becomes unhealthy or a user requests an invalid path, traffic should be redirected appropriately to the homepage or to a healthy server in order to maintain a smooth user experience and reduce service disruption.

Given the unpredictable nature of traffic growth, particularly during major news events, an effective Auto Scaling strategy is also required to automatically adjust capacity in response to changing demand.

Based on these requirements, the task is to evaluate and compare different server and instance options, analyze their performance and cost implications, and justify the most suitable choice. In addition, the available pricing models should be explained, followed by a recommendation for the option that best fits this use case.

Furthermore, a secure and scalable cloud network architecture should be designed in accordance with best practices. The proposed design should ensure proper resource isolation, controlled communication between components, minimal exposure of internal services, and efficient scalability as application demand increases.

Storage and Data Management

Regarding news-related media, including article images, event images, and thumbnails, all files are currently stored on the same web servers. This approach has led to storage capacity reaching its limits, requiring continuous manual management. In addition, this setup negatively impacts overall server performance and system stability.

Due to the nature of the business, data volume is increasing rapidly, with several gigabytes being added on a daily basis. Usage patterns indicate that most images are actively accessed within the first 30 days, after which access frequency declines significantly. Beyond 90 days, the data is primarily retained for archival purposes. Furthermore, to optimize storage costs, there is a requirement to remove outdated data, particularly previous file versions, after approximately 45 days.

Based on these storage requirements and data access patterns, an appropriate storage solution must be evaluated and selected. The chosen solution should be justified in terms of performance, scalability, cost efficiency, and data protection capabilities.

A recent incident highlighted the need for stronger data protection measures, where critical images associated with older articles were permanently deleted and could not be recovered. Therefore, the proposed solution must include mechanisms for data recovery in case of accidental deletion or modification, along with an additional layer of protection to prevent permanent deletion of sensitive data unless it is performed through a secure and controlled process.

Monitoring, Alerting, and Operational Visibility

Enhanced operational visibility across the environment is a key requirement. Notifications should be triggered whenever new instances are launched or existing instances are terminated as part of the Auto Scaling process. This ensures continuous tracking of scaling activities during both normal operation and traffic spikes.

In addition, full visibility into critical events within the storage environment is required, particularly those affecting the S3 bucket and its contents. Monitoring these activities enables better control over data and ensures that any changes are tracked and auditable.

Furthermore, proactive alerting is necessary to maintain system reliability. If the number of running instances approaches the maximum capacity defined in the Auto Scaling Group, an email notification should be generated as an early warning. This allows timely review of the system state and enables corrective actions before service performance or availability is impacted.

DNS and Application Security

The platform currently uses a domain name registered with Namecheap, and users must be able to access the application through it in a reliable and secure manner. Since the migration and application updates will be introduced gradually, an appropriate DNS routing strategy is required to direct a controlled percentage of traffic to a new application environment while maintaining the remaining traffic on the current environment. This approach enables updates to be tested safely under real production traffic, supports gradual adoption, and reduces the risk of service disruption during deployment.

Based on these requirements, an appropriate routing strategy should be implemented using Amazon Route 53. In addition, the role of AWS Certificate Manager (ACM) in securing the application should be explained, particularly in relation to enabling encrypted communication and supporting secure access through HTTPS.

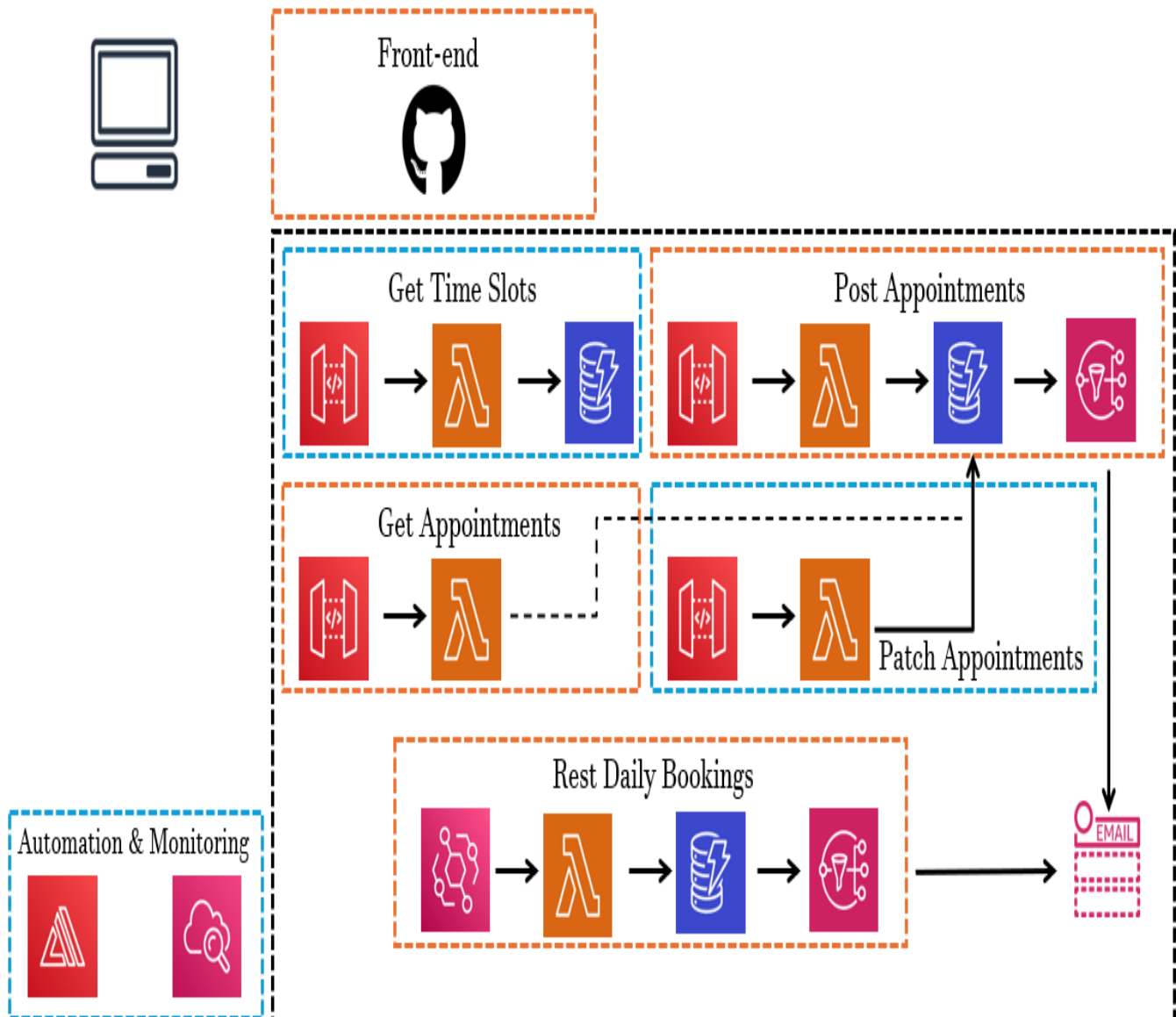
Scenario 2: HealthBooking Serverless System

This is Yazan from HealthBooking, a platform currently being developed to provide a web-based medical appointment booking system. The platform is intended to allow patients to reserve available time slots, while enabling administrators to manage appointments through a dedicated dashboard. In this phase of the project, the objective is to design and deploy the system using a cloud-based serverless architecture rather than relying on traditional server-based infrastructure.

The application consists of two main components: a frontend web application and a backend powered by AWS-managed services. The frontend should be hosted in a way that supports simple and efficient deployment directly from a GitHub repository. The backend is expected to expose APIs that support the core platform operations, including appointment creation, retrieval of available time slots, listing all appointments, and updating appointment status. In addition, the system must send notifications when new bookings are created and when daily bookings are reset.

The current solution is expected to make use of AWS Lambda, Amazon API Gateway, Amazon DynamoDB, Amazon SNS, Amazon EventBridge, AWS CloudWatch, and AWS Amplify.

The following diagram presents the overall system architecture and illustrates how the different components interact. It should be analyzed and used as a reference when designing the proposed solution.



Data Design

From a data perspective, the system is designed around two primary data entities. The first is the **Appointments** table, which stores essential booking information, including appointment ID, creation timestamp, patient name, selected time slot, status, and symptoms. The second is the **Slots** table, which maintains the list of available time slots along with their booking status.

The proposed data structure is presented below.

Attribute	Type	Description
appointmentId	String	Partition Key (UUID)
createdAt	String	Timestamp of booking
patientName	String	Patient name
slot	String	Time slot
status	String	Pending / In Progress / Completed
symptoms	String	Patient's symptoms

Table 3: Appointments Table

Attribute	Type	Description
slot	String	Partition Key
isBooked	Boolean	Whether slot is booked

Table 4: Slots Table

Based on this structure, HTUNews need you to evaluate and justify the database design and explain why it is suitable for this system.

Application Logic

From an application logic perspective, the platform is required to support several core operations, including retrieving available time slots, creating appointments, listing bookings for administrative purposes, updating appointment status, and performing a daily reset of bookings.

The backend implementation should follow a serverless architecture, where each operation is handled by a dedicated function. These functions must be exposed through well-defined API endpoints and seamlessly integrated with the frontend application.

The required API operations are outlined below.

Function Name	Method	Description
Get Time Slots	GET	Retrieves all available (unbooked) time slots.
Create Appointment	POST	Creates a new appointment, updates slot, and sends notification.
Get Appointments	GET	Returns all appointments for admin panel.
Update Appointment Status	PATCH	Updates appointment status.
Reset Daily Bookings	Scheduled	Clears bookings, resets slots, and sends notification.

Table 4: Backend Operations

Each function should be implemented as a separate serverless function and properly integrated with API endpoints. The API paths must be consistent with the frontend application.

Automation and Scheduling

The system is also required to perform automated daily maintenance. At 5:00 PM UTC each day, the platform should execute the following actions:

- Delete all appointments created during the current day
- Reset all time slots to an available state
- Send a notification confirming that the daily reset has been completed

Based on these requirements, explain how an event-driven architecture can be designed and used to implement this functionality efficiently and reliably.

Monitoring

At this stage, proper monitoring must be implemented to ensure that the platform operates as expected under different conditions. Visibility is required into key metrics such as system performance, resource utilization, and application behavior.

Accordingly, a monitoring solution should be designed and implemented using AWS CloudWatch, including a dashboard that presents a clear and well-organized view of the overall system status. The dashboard should make it possible to track critical metrics, identify potential issues, and support operational decision-making during periods of high traffic or service disruption.

(Deployment + CDN)

In this phase, the frontend application should be deployed using a continuous deployment approach from a GitHub repository in a simple and manageable manner. The deployment process must support automatic updates whenever changes are pushed to the repository, ensuring that the latest version of the application is consistently available to users.

In addition, the application must be accessible to users across different geographic locations with low latency and high performance. The proposed solution should enable efficient global content delivery and optimize the overall user experience regardless of user location.

Based on these requirements, explain how the selected deployment approach supports global application delivery, and how cloud networking principles contribute to improving performance, scalability, and user experience.

Learning Outcomes and Assessment Criteria			
Learning Outcome	Pass	Merit	Distinction
LO1 Understand Cloud Concepts and Global Infrastructure	P1 Explain cloud computing concepts, benefits, and service models. P2 Describe cloud global infrastructure (Regions, Availability Zones, Edge Locations).	M1 Compare and analyze the differences between traditional IT infrastructure, virtualization, and containerization, evaluating their impact on performance, scalability, and resource efficiency.	D1 Analyze the impact of using cloud architecture on availability, scalability, and fault tolerance.
LO2 Manage cloud Identity, Security, and Networking	P3 Explain Identity and Access Management (IAM) concepts P4 Describe networking concepts such as VPC, subnets, routing, and security groups	M2 Apply DNS routing strategies using Route 53 and explain the use of AWS Certificate Manager (ACM) for securing applications.	D2 Design secure and scalable cloud network architectures following best practices.
LO3 Work with Core cloud Compute, Storage, and Database Services	P5 Describe cloud computing services such as instances, Lambda, and basic container concepts. P6 Identify database and machine learning services in cloud environments	M3 Compare server-based and serverless architectures in terms of scalability, cost, and performance	D3 Evaluate and select appropriate storage and advanced database services (S3, EBS, RDS, DynamoDB) based on system requirements.
LO4 Integrate and Build Cloud Applications	P7 Describe application integration services such as API Gateway, SNS, SQS, and event-driven systems	M4 Explain how global application delivery is achieved using Software-Defined Networking (SDN) principles	
LO5 Deploy, Optimize, and Operate Cloud Systems	P8 Describe deployment methods, monitoring concepts and billing schemes in cloud environments	M5 Apply monitoring and alerting to maintain system performance and reliability	D4 Design automated cloud operations including deployment, scaling, and recovery